Nonlinear Equations

Alexandra A. Gomes

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Nonlinear Equations

Methods for solving

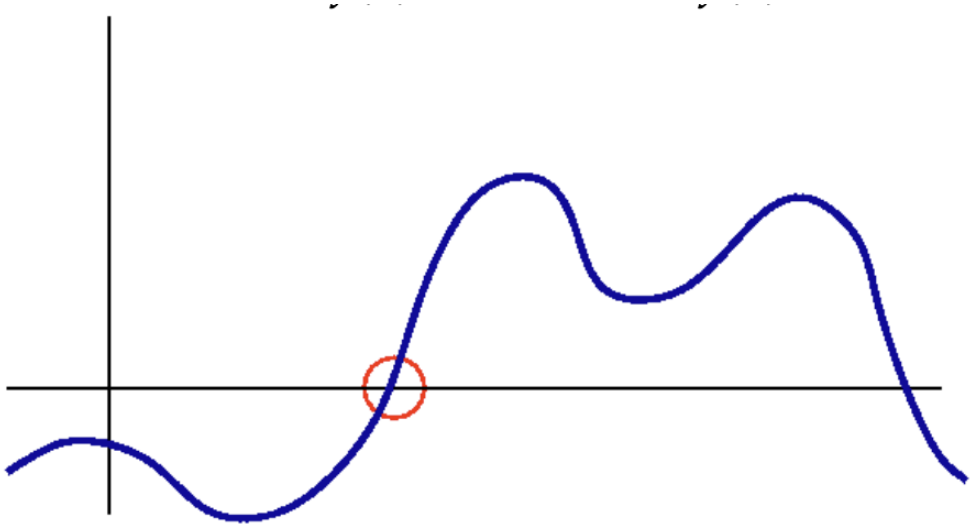
1. a single variable nonlinear equation;
2. systems of nonlinear equations;
3. nonlinear least squares problems.

Single Variable Nonlinear Equation

Our goal is to solve

$$f(x) = 0$$

to find x, which is a **zero** of f or a **root** of the equation.



Properties of Nonlinear Equations

- the unknown(s) appear *nonlinearly* in the equations;
- the canonical form of a nonlinear equation is $f(x) = 0$;
- how many roots? it depends on the function and on the interval of interest; for example:
  - $ax^2 + bx + c = 0$ can have two roots, one root, or no roots;
  - $\sin(x) = 0$ can have no root or many roots depending on the interval;
- in general, and unlike the linear case, we do not have direct methods that can find zeros of f;
- even roots of univariate polynomials (beyond quadratic) do not have simple closed form expressions; and no closed form for polynomials of degree higher than four;
- iterative methods are the only choice for general functions.

The Bisection Method

- the method is based on the Intermediate Value Theorem: let $f \in C[a, b]$ and K a number between $f(a)$ and $f(b)$; then, there exists a number $c \in (a, b)$ such that

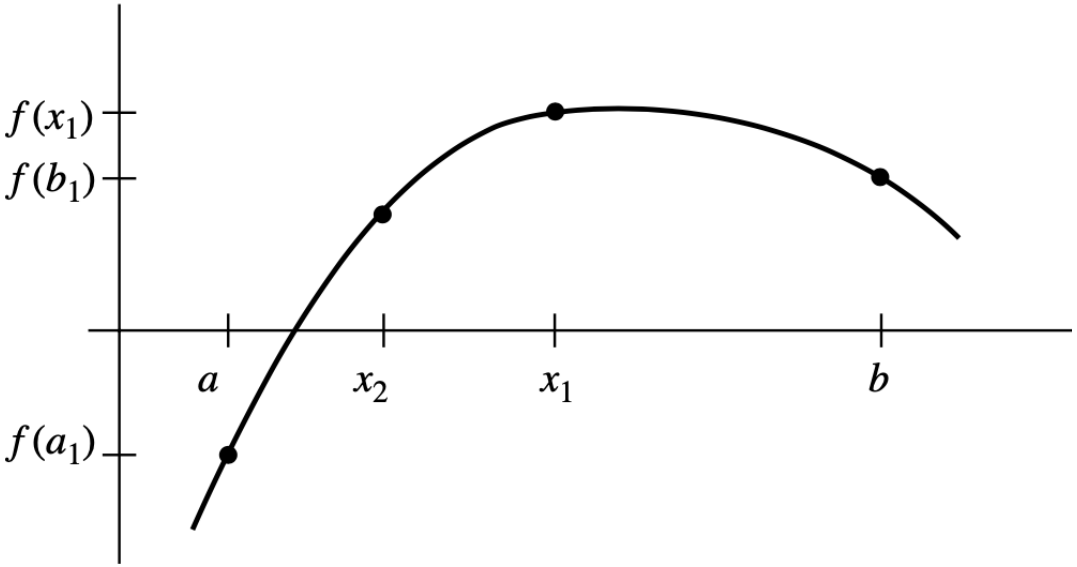
$$f(c) = K;$$

- choose $K = 0$ and a and b such that

$$f(a) f(b) < 0;$$

- main advantage: very robust;
- main drawback: low rate of convergence.

The Bisection Method Illustrated



Example

Suppose we want to determine $\sqrt{K}$, $K > 0$, which is equivalent to solving $f(x) = x^2 - K = 0$. With a and b such that $f(a)f(b) < 0$:

```
while abs(b − a) < ϵ
    x = (a + b)/2;
    if sign(x^2 − K) == sign(b^2 − K)
        b = x;
    else
        a = x;
    end
end
```

Root Bracketing

- the starting interval $[a, b]$ must contain the root: root bracketing;
- the starting interval $[a, b]$ is often called the uncertainty interval;
- each iteration reduces the uncertainty interval by $1/2$; hence the following result: suppose $f \in C[a, b]$ and $f(a) f(b) < 0$; the Bisection Method generates a sequence $\{c_n\}_{n=1}^{\infty}$ approximating a zero r of f with an absolute error satisfying

$$|c_n - r| \leq \frac{b-a}{2^n}, \quad n \geq 1;$$

- the starting interval $[a, b]$ should be as small as possible.

Why is this theorem useful?

Example

Determine the number of iterations necessary to find the root $f(x) = x^3 + 4x^2 - 10 = 0$, in $[1, 2]$, with accuracy 10^{-3}.

Find integer N that satisfies

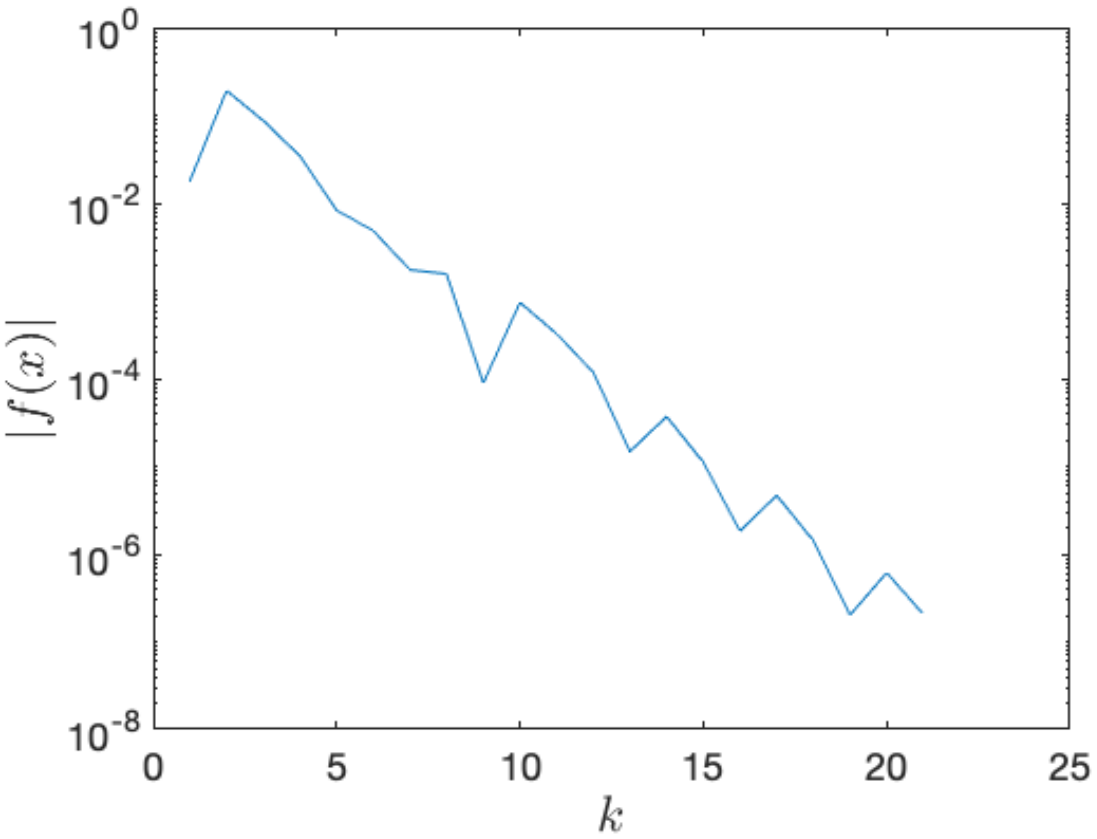
$$|c_N - r| \leq \frac{b-a}{2^N} = 2^{-N} \leq 10^{-3}.$$

The Bisection Method: Example

- consider
$$f(x) = x - x^{1/3} - 2 = 0;$$
- initial bracket: $[3, 4]$ with
$$f(a) = f(3) \approx -0.4422 \qquad f(b) = f(4) \approx 0.4126$$

k	a	b	x_{mid}	$f(x_{mid})$
0	3	4		
1	3	4	3.5	-0.01829449
2	3.5	4	3.75	0.19638375
3	3.5	3.75	3.625	0.08884159
4	3.5	3.625	3.5625	0.03522131
5	3.5	3.5625	3.53125	0.00845016
6	3.5	3.53125	3.515625	-0.00492550
7	3.51625	3.53125	3.5234375	0.00176150
8	3.51625	3.5234375	3.51953125	-0.00158221
9	3.51953125	3.5234375	3.52148438	0.00008959
10	3.51953125	3.52148438	3.52050781	-0.00074632

The Bisection Method: Example



Matlab Implementation

```
f1 = @(x) x - x.^(1/3) - 2;
a = 3;
b = 4;
niter = 10;
hist = zeros(niter+1,1);%history of iterates
for k = 1:niter
        mid = (a+b)/2;
        hist(k) = mid;
        if f1(a) * f1(mid) >= 0
                a = mid;
        else
                b = mid;
        end
end
```

Matlab Implementation

```
x = (a+b)/2;
hist(niter+1) = mid;
e = abs(f1(hist));      % error in f(x)
semilogy(1:niter+1, e)
```

The Bisection Method: Convergence Criteria

An automatic root-finding procedure needs to monitor progress towards the root and then stop when the current guess is close enough to the desired root. Two possible convergence checks are:

- check the closeness of successive approximations, for tolerance δ_x, which may be absolute or relative:

$$\left|x^{(k)} - x^{(k-1)}\right| < \delta_{x,\text{abs}} \quad \text{or} \quad \left|\frac{x^{(k)} - x^{(k-1)})}{b-a}\right| < \delta_{x,\text{rel}};$$

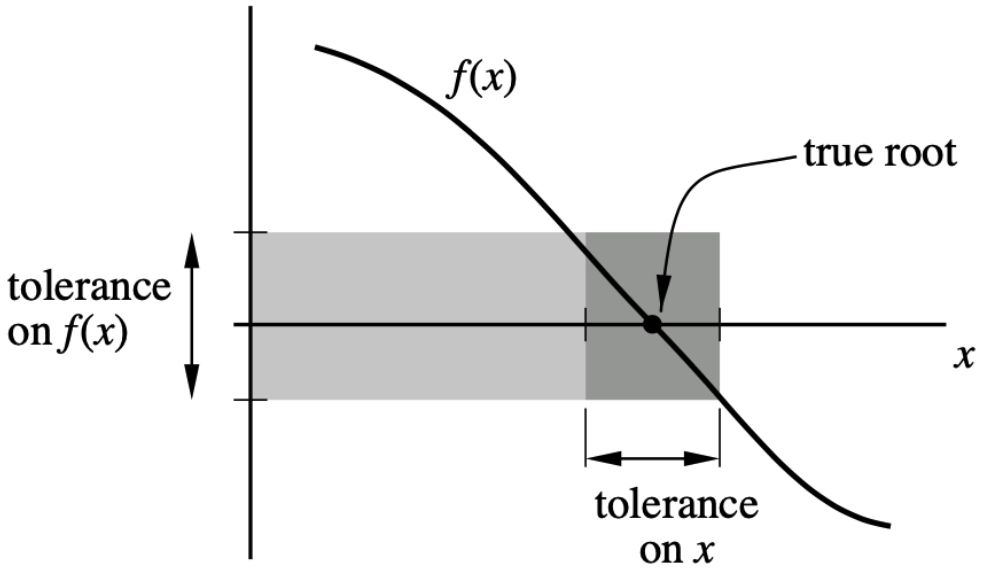
- check how close $f(x)$ is to zero at current guess for tolerance δ_f, absolute or relative:

$$\left|f(x^{(k)})\right| < \delta_{f,\text{abs}} \quad \text{or} \quad \frac{\left|f(x^{(k)}\right|}{\max(|f(a)|, |f(b)|)} \leq \delta_{f,\text{rel}}.$$

+ upper bound for the number of iterations

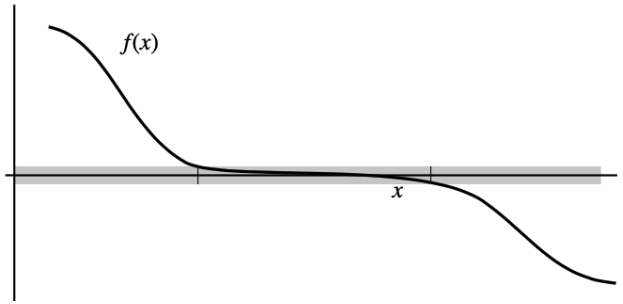
The Bisection Method: Which Criterion?

- the best criterion to use depends on the problem:

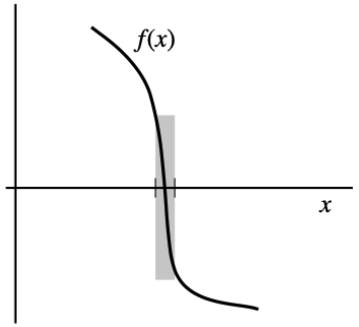


The Bisection Method: Convergence Criteria Compared

- if $f'(x)$ is small near the root, it is easy to satisfy tolerance on $f(x)$ for a large range of Δx;



  then, the tolerance on Δx is a conservative (safer) choice;
- if $f'(x)$ is large near the root, it is possible to satisfy the tolerance on Δx when $|f(x)|$ is still large:



  in this case, the tolerance on $f(x)$ is more conservative.

The Bisection Method: Relationship Between Criteria

- how are the criteria based on x and $f(x)$ related?
- consider the ratio:

$$\frac{f(x^{(k)}) - f(x^{(k-1)})}{x^{(k)} - x^{(k-1)}}$$

- in the limit, as $x^{(k-1)}$ and $x^{(k)}$ converge to the exact answer r is just $f'(r)$;
- we can thus expect that

$$\left| f(x^{(k)}) - f(x^{(k-1)}) \right| \approx \left| f'(r) \right| \left| x^{(k)} - x^{(k-1)} \right|$$

as $x^{(k)}$ approaches the solution r.

Matlab Implementation

```
function [sol, history] = bisect(f, a, b, tol)
% call is [x_star, hist] = bisect(f1, 3, 4, 1e-6)
history = [];
mid = (a+b)/2;
while abs(f(mid)) > tol
        history = [history; mid];
        if f(a) * f(mid) >= 0
                a = mid;
        else
                b = mid;
        end
mid = (a+b)/2;
end
sol = mid;
history = [history; sol];
end
```

Rate of Convergence

We already worked with the term of *rate of convergence* twice:

1. in the first 2 weeks, preliminaries (algorithms.pdf): we compared the error at iterate k ($\left|x^{(k)} - r\right|$) with the error of a know sequence of iterates;
2. in iterative methods for systems of linear equations (Iterative LA 2.pdf): we answered the question of how many iterations until the error norm dropped an order of magnitude from the initial error norm;

Here, we will compare the error at iterate $k+1$ with the error at iterate k, so we are following the same definition as in 2.

Rate of Convergence

A sequence of iterates is said to converge with rate of convergence r if

$$\left|e^{(k+1)}\right| \leq C \left|e^{(k)}\right|^r$$

in the limit $k \to \infty$.

Convergence is

linear if $r = 1$ and $C < 1$; linear convergence adds a constant amount of accuracy at each step;

superlinear if there is a sequence $\rho_k \to 0$, such that

$$\left|e^{(k+1)}\right| \leq \rho_k \left|e^{(k)}\right|;$$

quadratic if $r = 2$; quadratic convergence adds double the number of digits of accuracy, relative to the previous step, at each step $(10^{-1}, 10^{-2}, 10^{-4}, 10^{-8}, 10^{-16})$;

Linear convergence

Linear convergence means that the number of correct digits grows by about the same fixed amount per iteration.

An example is

$$x_k = 1 + 2^{-k},$$

which converges to 1. Its error is

$$e_k = |x_k - 1| = 2^{-k}.$$

Hence,

$$e_{k+1} = 2^{-(k+1)} = \frac{1}{2}\,2^{-k} = \frac{1}{2}e_k.$$

So at each step the error is multiplied by the same constant factor $1/2$.

Linear convergence and decimal accuracy

The number of correct digits is approximately

$$-\log_{10}(e_k).$$

If

$$e_{k+1} = C\, e_k, \qquad 0 < C < 1,$$

then

$$-\log_{10}(e_{k+1}) = -\log_{10}(C\, e_k) = -\log_{10}(e_k) - \log_{10}(C).$$

Since $-\log_{10}(C)$ is a constant, the accuracy increases by the same fixed amount at every step.

For $e_{k+1} = \frac{1}{2} e_k$,

$$-\log_{10}(e_{k+1}) = -\log_{10}(e_k) - \log_{10}\left(\frac{1}{2}\right) = -\log_{10}(e_k) + \log_{10}(2).$$

Thus each iteration adds $\log_{10}(2) \approx 0.301$ decimal digits of accuracy.

The Bisection Method: Rate of Convergence

For the error in the iterates in x:

- in the worst case, we have

$$\left|r - x^{(k+1)}\right| = \left|e^{(k+1)}\right| = \tfrac{1}{2}\left|e^{(k)}\right|;$$

- and therefore, the bisection method converges *linearly*;
- also:

$$\left|e^{(k)}\right| = \left(\tfrac{1}{2}\right)^k \left|e^{(0)}\right|,$$

  which on a semilogy plot results in the line

$$\log\left|e^{(k)}\right| = \log(\tfrac{1}{2})\, k + \log\left|e^{(0)}\right|$$

- the error in the iterates in $f(x)$ behave similarly, as per our previous discussion.

Goal: Better rate of Convergence

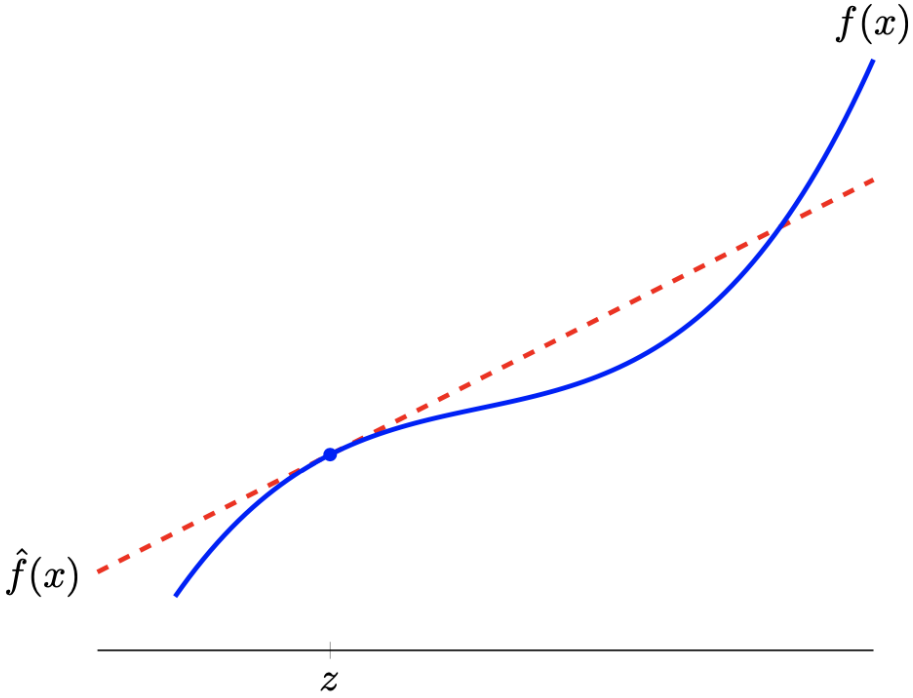
- we would like a better convergence rate;
- the bisection method evaluates f once per iteration;
- if we used more information from f, could we improve convergence?

Food for Thought

What if we use the bisection method to find the roots of

$$x^2 = 0?$$

Linear Approximation



Background: linear Taylor approximation

- the first-order (linear) Taylor approximation about z is

$$\hat{f}(x) = f(z) + f'(z)(x - z);$$

- the error in this approximation is

$$|f(x) - \hat{f}(x)| < \tfrac{1}{2} M (x - z)^2,$$

$$M = \max_{z<\xi<x} f''(\xi)$$

- sometimes, $\hat{f}$ is written as

$$\hat{f}(x; z) = f(z) + f'(z)(x - z).$$

Newton's method

- core idea: replace f by a linear Taylor approximation $\hat{f}$ centered at $x^{(k)}$:

$$f(x) \approx \hat{f}(x; x^{(k)}) = f(x^{(k)}) + f'(x^{(k)})(x - x^{(k)});$$

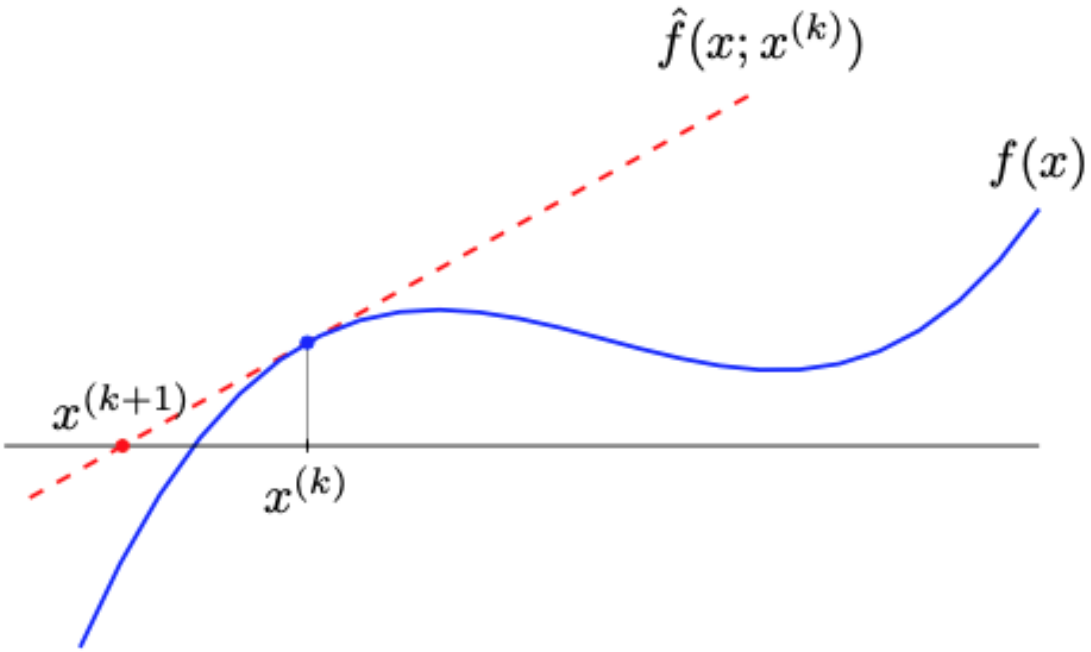
- set $x^{(k+1)}$ as the solution of $\hat{f}(x; x^{(k)}) = 0$:

$$f(x^{(k)}) + f'(x^{(k)})(x^{(k+1)} - x^{(k)}) = 0;$$

- if $f'(x^{(k)}) \neq 0$,

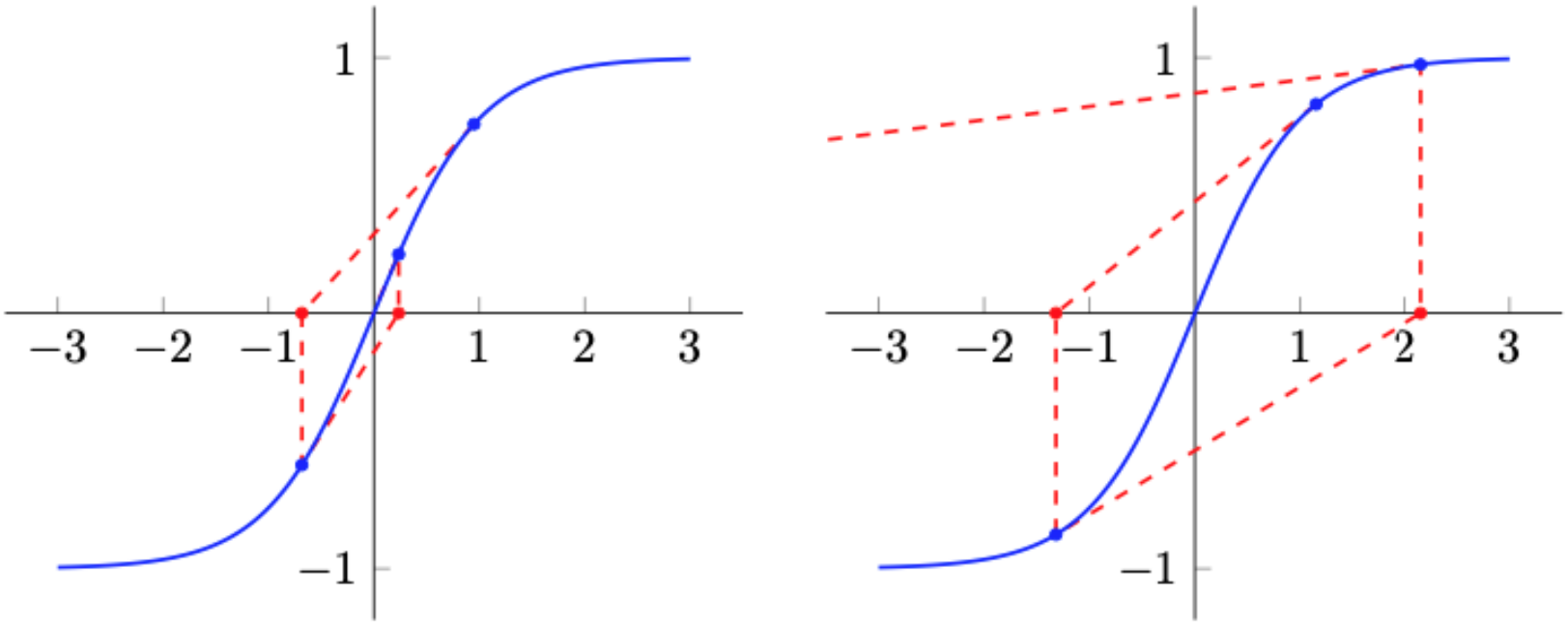
$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}.$$

Newton's Method Illustrated



Does Newton's Method Always Converge?

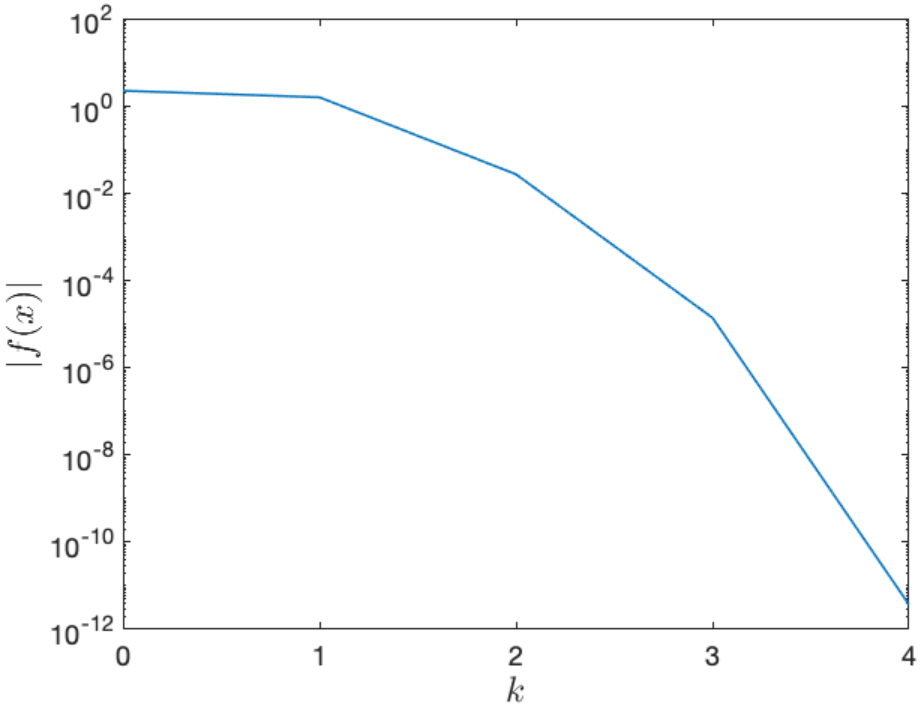
- Newton's method converges when the starting point $x^{(0)}$ is "sufficiently close" to r:



- Newton's method fails if it encounters a point $x^{(k)}$ where $f'(x^{(k)}) = 0$;
- yet, it is the main workhorse for solving nonlinear equations!

Newton's Method: Rate of Convergence

- let f have continuous f'' in $[a, b]$ where there is a simple root r ($f'(r) \neq 0$);
- then there is a Δ such that starting with $x^{(0)}$ anywhere in the interval $[r - \Delta, r + \Delta]$, Newton's method converges *quadratically*:



Matlab Implementation

```
function [sol, his] = newton(f, fprime, x0, tol)
history = [];
x = x0;
while abs(f(x)) > tol
his = [his; x];
x = x - f(x) / fprime(x);
end
sol = x;
his = [his; sol];
end
```

example:

```
f1 = @(x) x - x.^(1/3) - 2;
f1p = @(x) 1 - (1/3) * x.^(-2/3);
[x, hist] = newton(f1, f1p, 0.5, 1e-12);
semilogy(0:length(h)-1, abs(f1(h))
```

More Food for Thought

What if we use Newton's method to find the roots of

$$x^2 = 0?$$

The Secant Method

- Newton's method needs f';
- the Secant Method circumvents the need for explicitly evaluating this derivative, by substituting f' by its finite difference approximation:

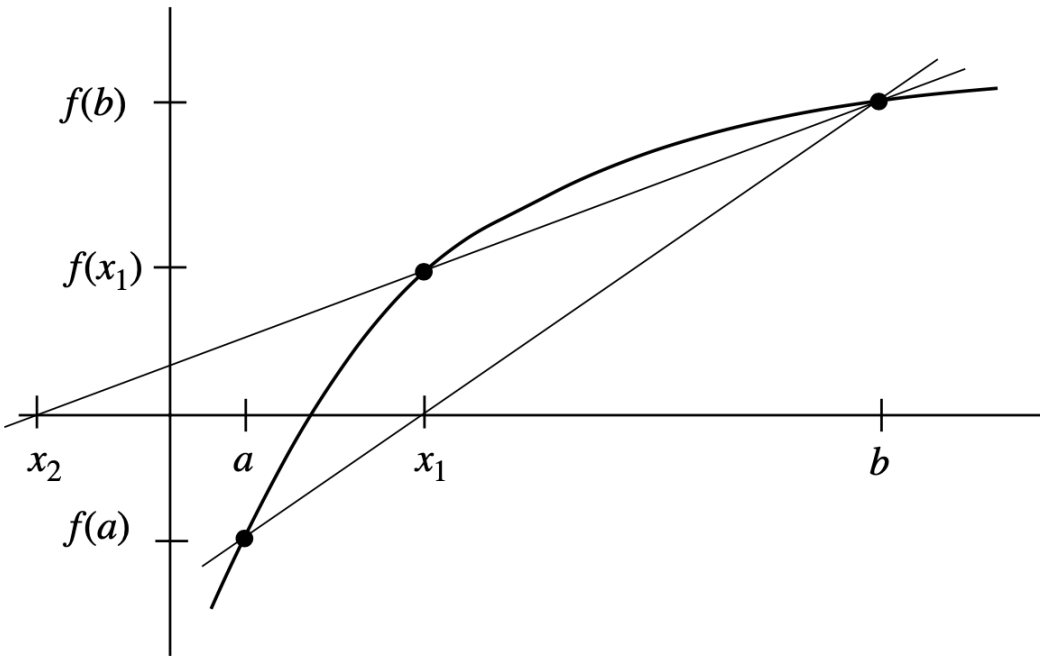
$$f'(x^{(k)}) \approx \frac{f(x^{(k)}) - f(x^{(k-1)})}{x^{(k)} - x^{(k-1)}}$$

- the secant iteration or quasi-Newton iteration is

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})\,(x^{(k)} - x^{(k-1)})}{f(x^{(k)}) - f(x^{(k-1)})};$$

- the secant method needs *two* starting iterates.

The Secant Method Illustrated

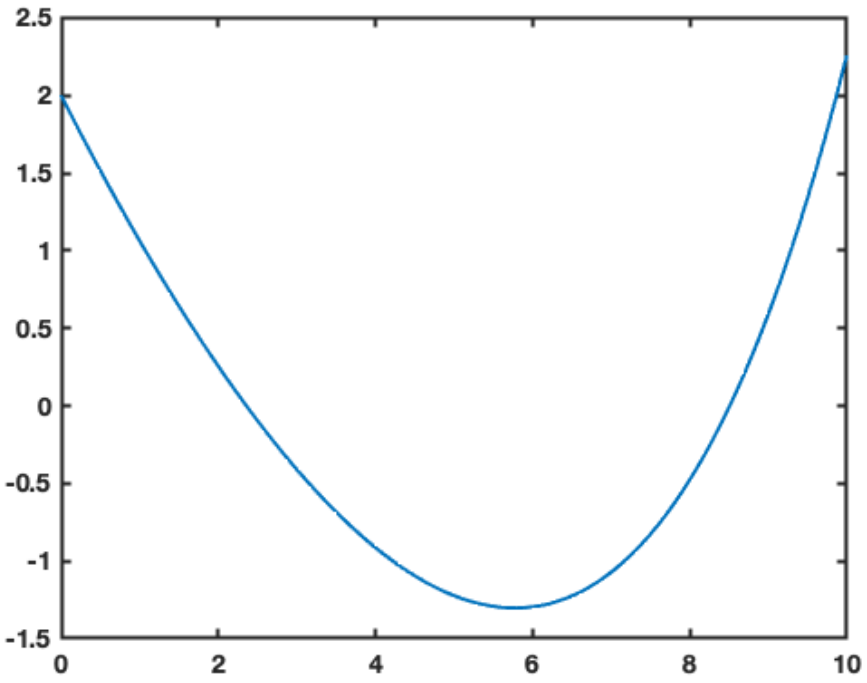


The Secant Method: Rate of Convergence

- as with Newton's method, there is no guarantee that the secant method converges;
- initial iterates must be "close" to the root;
- superlinear convergence with $r \approx 1.6$;
- much faster than the bisection method but not quite as fast as the Newton's iteration;
- variations on the basic idea can make it more robust but will slow down the convergence to linear rate.

Comparison

$$\exp(x/4) + \exp(-x/4) - x = 0$$



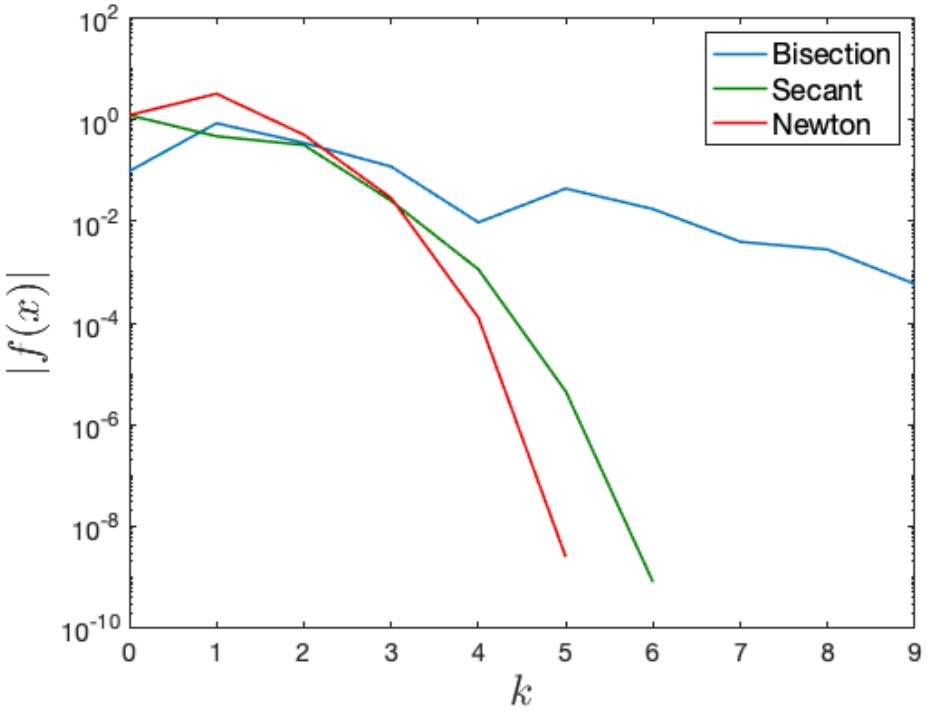
Comparison

```
f2 = @(x) exp(x/4) + exp(-x/4) - x;
f2p = @(x) 0.25*exp(x/4) - 0.25*exp(-x/4) - 1;

[x1, h1] = bisect(f2, 0, 5, 1e-3);
% needs the interval [0, 5]
[x2, h2] = secant(f2, 0, 5, 1e-8);
% needs 2 starting points 0, 5
[x3, h3] = newton(f2, f2p, 5, 1e-8);
% needs a starting point and f'

semilogy(abs(f2(h1))
hold on
semilogy(abs(f2(h2))
semilogy(abs(f2(h3))
```

Comparison


Bisection
Secant
Newton
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
$|f(x)|$
0
1
2
3
4
5
6
7
8
9
k

Fixed Point Iteration

A fixed point of a function g is a point $x^\star$ satisfying

$$x^\star = g\left(x^\star\right).$$

The idea is that $f(x) = 0$ can be written as $x = g(x)$ and that

$$x^{(k+1)} = g(x^{(k)}), \quad k = 0, 1, 2, \ldots$$

with a given $x^{(0)}$ converges to a fixed point. This algorithm is called the Fixed Point Iteration.

Existence and Uniqueness of a Fixed Point

1. if $g \in C[a, b], g(a) \geq a, g(b) \leq b$, then there exists a fixed point $x^\star \in [a, b]$; the proof is based on the Intermediate Value Theorem;
2. in addition, if g' exists and there exists $\rho < 1$ such that

$$|g'(x)| \leq \rho,$$

for all $x \in (a, b)$, then $x^\star$ is unique in $[a, b]$; the proof is by contradiction and based on Taylor truncation error.

Convergence of the Fixed Point Iteration

1. question: under the same assumptions, does the fixed point iteration converge to $x^\star$?
2. $|x^{(k+1)} - x^\star| = |g(x^{(k)}) - g(x^\star)| = |g'(\xi)(x^{(k)} - x^\star)|$, with ξ between $x^{(k)}$ and $x^\star$;
3. then
$$|x^{(k+1)} - x^\star| \leq \rho |x^{(k)} - x^\star|,$$
which represents a contraction by the factor ρ;
4. moreover,
$$|x^{(k+1)} - x^\star| \leq \rho^{k+1} |x^{(0)} - x^\star|;$$
5. as $k \to +\infty$, $\rho^k \to 0$, establishing convergence of the fixed point iteration.

Rate of Convergence of the Fixed Point Iteration

$$|x^{(k+1)} - x^\star| \leq \rho |x^{(k)} - x^\star|$$

1. linear rate of convergence;
2. contrary to the bisection method, where $C = 1/2$ does not depend on f, ρ depends on $|g'|$ and hence on f;
3. how many iterations to reduce the error by a factor of 10, an order of magnitude?

$$|x^{(k)} - x^\star| \approx \rho^k |x^{(0)} - x^\star| \approx 0.1 |x^{(0)} - x^\star|$$

$$k = -\frac{1}{\log_{10} \rho}.$$

4. note that the convergence properties of the iteration depend on the choice of g.

Zeros of Polynomials: an Eigenvalue Problem

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

- the eigenvalues of A and the zeros of its characteristic polynomial $p(\lambda)$ are the same:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -c_0 & -c_1 & -c_2 & -c_3 \end{bmatrix},$$

$$p(\lambda) = \lambda^4 + c_3\lambda^3 + c_2\lambda^2 + c_1\lambda + c_0;$$

- we can find all the roots and we don't need to iterate; but only applicable for polynomials.